Solved

Tutorial examination

Course-335, Time 50 mins, Full marks: 20

Answer all the following questions:

1. Why is metallic Na used in fusion of organic compounds? Is it possible to replace Na by Li or K? Write the formula of possible compounds formed upon decomposition with Na. 3.0
2. What is stock solution? How halides ions cane be differentiate from stock solution? 2.5
3. An organic compound is insoluble in 5% NaOH, 5% HCl and *Conc*. H_2SO_4. If there is no nitrogen or sulphhur present, assume the solubility class of the compound. Mention the name of probable compounds of this class. 2.5
4. An organic compound shows the IR bands at 3385 and 1685 cm^{-1} It produce brisk effervescence upon addition of *dil* HCl and 20% $NaNO_2$ rather reacting with 2,4-DNP to form red dye. Assume the class of the compound and give necessary explanation. 3.0
5. What is meant by derivative? How it is useful in identification of unknown organic compounds
6. Write the approximate value of ν_{N-H}. How can you differentiate primary, secondary and tertiary amines by IR spectroscopy? 2.0
7. Elucidate the structure of a compound having the molecular formula $C_8H_{11}N$ with the help of following spectroscopic data. 3.0

   IR: υ = 3450, 3300, 3100, 2900, 2850, 1600, 1510 cm^{-1}

   NMR: δ = 1.2(3H, t), 2.35(2H, q), 3.5(2H, br, s), 6.5-7.3(4H, m) ppm.

michelines

Organic Lab

1. Metallic Na is used in fusion of organic compound because of electropositive nature & in order to covert covalent form to ionic form. It is quite reactive in nature.

K is more reactive than Na and it reacts violently on heat. Again if any organic compound contains nitrogen then on fusion with K it produce KCN which is very poisionous. So, K is not used instead of Na. Li is not used in fusion because it is less reactive than Na and hence requires time to complete fusion.

$$C, N, S, X \xrightarrow{\text{heat}} Na_2S + NaCN + NaX \quad [X = Cl, Br \text{ and } I]$$

2. stock solution:

Halides ions (Cl⁻, Br⁻ and I⁻) can be differentiated from the stock solution by treating it with ccl4 and conc. HNO_3 and then shaking vigorously.

(i) Iodine will produce violet colour in the ccl4 layer.

$$NaI \xrightarrow[ccl_4]{HNO_3} NaNO_3 + HI$$

$$HI + HNO_3 \longrightarrow I_2 \text{ (in ccl4)}$$
(violet colour)

(ii) Bromine will produce deep brown or tan or tan red colour in the ccl4 layer.

$$NaBr \xrightarrow[ccl_4]{HNO_3} NaNO_3 + HBr$$

$$HBr + HNO_3 \longrightarrow Br_2 \text{ (in ccl4)}$$
(tan red / deep brown colour)

(iii) Chlorine will produce no colour in ccl4 layer.

$$NaCl \xrightarrow[ccl_4]{HNO_3} NaNO_3 + HCl$$

$$HCl + HNO_3 \longrightarrow Cl_2 \text{ (in ccl4)}$$
(colourless)

3. An organic compound is insoluble in 5% NaOH, 5% HCl and conc. H_2SO_4 and there is no nitrogen or sulphur present. So, the solubility class of the compound is Class I (Inert).

Name of the probable compounds of this class is: Saturated hydrocarbons, alkyl halides, aryl halides, deactivated aromatic compounds, diaryl ethers etc.

4. Given, the organic compound shows the IR bands at 3385 and 1685 cm^{-1}.

IR value (cm^{-1}):	Inferences
3385	Indicates N–H stretching vibration
1685	Indicates $-\overset{O}{\overset{\|}{C}}-$ stretching vibration of amide

The products produce brisk effervescence upon addition of dil. HCl and 20% $NaNO_2$

rather reacting with 2,4-DNP to form Red dye.

So, the compound is amide.

The compound belongs to the MN class.

~~Because~~

Class MN:

Miscellaneous neutral compounds containing nitrogen or sulphure and having more than five carbon.

Nitrogen present: Anilidides.

Amides are insoluble in H_2O, 5% NaOH, Insoluble in 5% HCl and contains nitrogen.

So, it belongs to the MN class.

Q. A derivative is a compound that is synthesised from an identical compound by a chemical reaction with the replacement of one atom or group of atoms.

Organic compound derivatives help to classify the specified compound with similare melting and boilling point. Organic compound derivatives have different melting or boilling points, so the specified compounds can easily be easily distinguished. for example: naphthalene and anthracene have the same boilling point and both are hydrocarbons, so they can be separated by the derivatives of both hydrocarbons.

CH_2 और Aromatic ring के साथ जुड़े हुए हाइड्रोजन परमाणु
20° (2.4-2.8) Bulk में और साथ ही N से जुड़े हुए हाइड्रोजन परमाणु 3 से
कम होते हैं।

6. The approximate value of ν_{N-H} is 3300 – 3500 cm^{-1}.

Primary amines have two N-H bonds, and therefore two N-H stretching peaks; secondary amines have one N-H bond and hence one N-H stretching peak and tertiary amines have no N-H bonds and hence no N-H stretching peaks.

7.

IR (cm^{-1})		Inference
3450, 3300	⟶	Indicates N-H stretching vibration
3100	⟶	sp^2 hybridized C-H stretching vibration
2900, 2850	⟶	sp^3 hybridized C-H stretching vibration
1600, 1510	⟶	Aromatic or olephinic C=C stretching vibration

NMR (ppm)		Inference
1.2 (3H, t)	⟶	Indicates terminal $-CH_3$ group
2.35 (2H, q)	⟶	Indicates $-CH_2-$ group
3.5 (2H, br, s)	⟶	Indicates $-NH_2-$ group
6.5 - 7.3 (4H, m)	⟶	Indicates disubstituted benz aromatic ring

∴ The probable structure of the compound is,

